

PES UNIVERSITY, BANGALORE

Department of Computer Science and Engineering

SOFTWARE ENGINEERING - LEVEL 3 MINI-PROJECT

SYNOPSIS / PROJECT PROPOSAL

TRAFFIC MANAGEMENT SYSTEM (SIMULATION)

TEAM 11

Project Detail	Information
Course	UE24CS341A – Software Engineering
Semester	5
Group Number	Team 11
Development Methodology	Agile
Technology	C / C++
Version	1.0
Academic Year	2026
Project ID	TBD

Team Members

Sl. No.	Student Name	SRN
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1. Project Overview

Traffic Management System (Simulation) is a C/C++ based software simulation that models vehicle movement through interconnected roads and intersections. The system will simulate traffic queues, traffic-signal operation, varying traffic densities, emergency vehicle priority and simulated traffic incidents. It will collect measurable traffic-performance metrics such as queue length, waiting time and throughput and use them to evaluate and compare traffic-management strategies.

2. Problem Statement

Conventional traffic signal operation often relies on predefined signal timings and may not respond efficiently to changing traffic conditions. Uneven traffic distribution can result in unnecessary queues and waiting time at intersections, while emergency vehicles may face delays when normal signal cycles are followed. Evaluating alternative traffic-management strategies directly on real roads is impractical and unsafe. The proposed system addresses this problem by providing a controlled simulation environment in which vehicle flow, traffic density, signal strategies, emergency priority and traffic incidents can be modelled and evaluated using measurable performance indicators.

3. Proposed Project Description

The proposed system will represent a configurable road network containing roads, lanes and intersections controlled by simulated traffic signals. Vehicles will be generated according to configurable traffic conditions and assigned movement paths through the network. Vehicles will enter queues when movement is restricted and will proceed when the corresponding signal permits movement.

The simulation will support both fixed-time and adaptive traffic signal strategies. In adaptive mode, the system will use traffic-density information to adjust signal operation. Simulated emergency vehicles will receive higher priority, and simulated road incidents can be introduced to study their effect on traffic flow. At the end of a run, the system will provide traffic statistics that can be used to compare management strategies.

4. Target Users / Stakeholders

User / Stakeholder	Role / Need
Simulation Operator	Configures traffic scenarios, starts/stops simulations and introduces simulation conditions.
Traffic Administrator	Configures road networks, intersections, signals and traffic-management strategies.
Traffic Analyst	Examines traffic metrics and compares the performance of different strategies.
Project Development Team	Develops, tests, validates and maintains the simulation software.

5. Objectives

- To develop a C/C++ based simulation of vehicle movement through interconnected roads and intersections.
- To simulate vehicle flow under different traffic-density conditions.
- To implement fixed-time traffic signal control for baseline comparison.
- To implement adaptive traffic signal control based on simulated traffic conditions.
- To reduce simulated vehicle waiting time through appropriate adaptive signal decisions.
- To provide priority to simulated emergency vehicles.
- To simulate road incidents and observe their effect on traffic flow.
- To measure traffic-performance indicators such as queue length, waiting time and throughput.
- To compare traffic-management strategies using measurable simulation results.
- To develop the system incrementally using Agile software-development practices.

6. Scope

6.1 In Scope

- Road, lane and intersection modelling.
- Traffic-signal configuration and simulation.
- Vehicle generation under configurable traffic-density conditions.
- Vehicle movement and queue formation.
- Fixed-time traffic-signal strategy.
- Adaptive traffic-signal strategy.
- Emergency vehicle priority handling.
- Simulated traffic incidents such as road blockages.
- Traffic monitoring and collection of simulation statistics.
- Comparison of traffic-management strategies.

6.2 Out of Scope

- Control of physical traffic-signal hardware.
- Real-time city traffic-data integration.
- GPS-based tracking of real vehicles.
- Integration with actual emergency services.
- Deployment for controlling traffic on public roads.
- Autonomous vehicle control.
- Direct integration with real-world traffic cameras or sensors.

7. Functional Features

ID	Functional Feature	Description
F1	Road & Intersection Management	Configure roads, lanes, intersections and traffic-signal configurations.
F2	Vehicle Generation & Traffic Flow	Generate vehicles under configurable traffic conditions and simulate their movement and queues.
F3	Traffic Signal Management	Operate traffic signals using defined fixed-time and adaptive strategies.
F4	Adaptive Traffic Control	Calculate traffic density and dynamically modify signal operation when adaptive mode is enabled.
F5	Emergency Vehicle Priority	Identify simulated emergency vehicles and provide priority through appropriate signal control.
F6	Traffic Incident Simulation	Introduce simulated road incidents and model their effect on traffic flow.
F7	Traffic Monitoring	Monitor traffic density, queue length, waiting time and vehicles processed.
F8	Performance Analytics	Generate simulation statistics and compare traffic-management strategies.

8. Team-wise Functional Ownership

Member	Primary Functional Ownership	Secondary Responsibilities
Rohan Komperla	Road & Intersection Management	Architecture support and GitHub coordination
Rujul Jain	Vehicle Generation & Traffic Flow	Integration and module interfaces
Shana Susan Varghese	Adaptive Traffic Signal Control	Algorithm validation and interface support
M Ajjaiah	Emergency Vehicle & Incident Management	Testing and CI/CD support

All team members will participate in requirements analysis, architecture/design, implementation, testing, validation, documentation and Agile project activities. The primary ownership above identifies the complete functionality for which each member will be responsible.

9. Plan of Work

Sprint	Planned Activities	Expected Output / Acceptance Evidence
Sprint 1 – Requirements & Setup	Finalize problem, scope, use cases and initial requirements; set up GitHub/Jira; establish C++/CMake project structure.	Approved synopsis/SRS baseline, repository, backlog and initial project structure.
Sprint 2 – Road Network & Vehicle Simulation	Implement road/intersection model, vehicle generation, queues and basic vehicle movement.	Working road-network and vehicle-flow simulation with unit tests.
Sprint 3 – Traffic Signal Control	Implement signal states, fixed-time control, traffic-density calculation and adaptive signal logic.	Working fixed/adaptive signal strategies with test evidence.
Sprint 4 – Emergency & Incident Management	Implement emergency vehicle priority and simulated traffic incidents.	Integrated emergency/incident functionality and validation evidence.
Sprint 5 – Analytics & Integration	Implement waiting-time, queue-length and throughput metrics; compare strategies; integrate modules.	Integrated simulation, performance results and integration tests.
Sprint 6 – Validation & Release	System testing, coverage, CI/CD, static analysis/security validation, documentation and demo preparation.	Final validation evidence, coverage report, release build and presentation-ready demo.

10. Proposed Technology and Tools

Activity / Area	Proposed Technology / Tool	Purpose
Programming	C++17	Implementation of simulation and traffic-management modules.
Build System	CMake	Build configuration and reproducible project builds.
Version Control	Git + GitHub	Source control, branches, commits and pull requests.
Agile Project Management	Jira	Backlog, user stories, tasks and sprint tracking.
Unit Testing	GoogleTest	Automated unit testing of core C++ modules.
CI/CD	Jenkins	Automated build and test pipeline.
Containerization	Docker	Reproducible development/build environment where applicable.
Static Analysis	SonarQube or similar	Code-quality and static-analysis checks.
UML / Design	Draw.io / PlantUML	Use-case and software-design diagrams.
Simulation Interface	C++ terminal or graphical interface	Display simulation state and traffic statistics.

11. Agile Development Approach

The project will follow an Agile development approach. The work will be divided into short iterations, with functionality represented through user stories and tasks assigned to team members. GitHub will be used to maintain source-code history and pull requests, while Jira will be used for backlog and sprint management. Features will be developed and integrated incrementally, with testing and validation performed throughout development rather than only at the end of the project.

12. Expected Outcomes

- A working C/C++ traffic-management simulation.
- A configurable road and intersection model.
- Simulation of vehicle generation, movement and queues.
- Fixed-time and adaptive traffic-signal strategies.
- Emergency vehicle priority handling.
- Simulation of selected traffic incidents.
- Collection of waiting-time, queue-length and throughput metrics.
- Comparison of traffic-management strategies using simulation results.
- Automated testing and supporting software-engineering evidence.

- A maintainable modular implementation suitable for later design, testing and validation phases.

13. Preliminary Success / Evaluation Criteria

- The system shall successfully execute configured traffic scenarios without abnormal termination.
- Vehicle movement and queue formation shall follow the configured road and signal rules.
- Traffic signals shall transition according to the selected control strategy.
- Adaptive control shall respond to simulated traffic-density changes.
- Emergency vehicles shall receive the specified priority behaviour.
- Traffic incidents shall affect the relevant simulated road conditions.
- The system shall calculate and report defined traffic-performance metrics.
- Fixed-time and adaptive strategies shall be comparable using the same traffic scenario.
- Core modules shall be supported by automated unit tests and project test evidence.

14. Preliminary Use Cases

Use Case ID	Use Case	Primary Actor	Expected Outcome
UC-01	Configure Road Network	Simulation Operator / Administrator	Roads, lanes and intersections are configured.
UC-02	Configure Traffic Signals	Simulation Operator / Administrator	Signal states and strategy parameters are configured.
UC-03	Start Simulation	Simulation Operator	Configured simulation begins execution.
UC-04	Generate Traffic	Simulation System	Vehicles are generated according to selected traffic conditions.
UC-05	Simulate Vehicle Movement	Simulation System	Vehicles move through roads and intersections subject to signal conditions.
UC-06	Manage Adaptive Signals	Traffic Management Module	Signal operation responds to simulated traffic density.
UC-07	Handle Emergency Vehicle	Simulation Operator / System	Emergency vehicle receives priority treatment.
UC-08	Simulate Traffic Incident	Simulation Operator	Incident affects the selected road/traffic conditions.
UC-09	Monitor Traffic	Traffic Analyst / Operator	Current traffic conditions and queues are displayed.
UC-10	View Simulation Analytics	Traffic Analyst	Performance metrics and strategy comparisons are displayed.

15. Initial Requirement-to-Feature Direction

The functional features and use cases in this synopsis will form the baseline for the Software Requirements Specification (SRS). During the requirements and design phases, each approved requirement will receive a unique identifier and will be traced through architecture, design, implementation and test cases. This ensures continuity across the later SDLC deliverables.

16. Conclusion

The proposed Traffic Management System (Simulation) will provide a controlled environment for modelling traffic flow and evaluating traffic-signal strategies. By combining vehicle-flow simulation, adaptive signal control, emergency priority, incident simulation and measurable performance analytics, the project provides a clear set of functional modules that can be developed and evaluated independently by the four team members. The modular C/C++ implementation will be developed incrementally using Agile practices and supported by testing, version control and continuous integration activities.

— End of Synopsis —